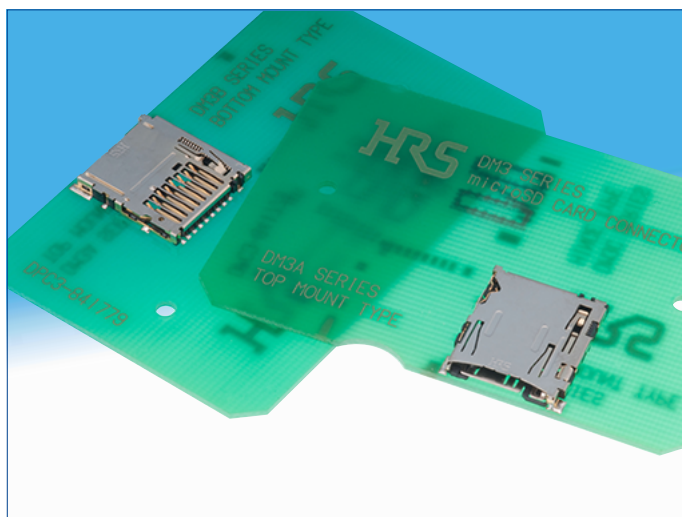


# microSD™ Card Connectors

## DM3 Series



### ■ Features

#### ◆ Common to the entire Series

##### 1. Extremely small in size

Small external dimensions and the above-the-board height make the connectors the smallest on the market.

##### 2. Reverse card insertion protection

Unique card slot design (patented) protects the connector from damage when the card is attempted to be inserted in reverse, allowing it to re-inserted correctly.

##### 3. Effective ground and shield configuration

4-connection points of the metal cover to the printed circuit board assures secure connection of the ground circuit and provides EMI protection.

##### 4. Rigid and strong construction

Despite its small size, high-strength materials used in the connectors produced a strong and rigid structure.

##### 5. Card detection switch

The card detection switch is Normally Open

#### ◆ DM3AT and DM3BT (Push - Push, with ejection mechanism)

##### • Card fall-out prevention

Built-in card tray and the unique push insertion-push ejection mechanism (patented) prevent accidental card ejection or fall-out.

Despite its small size the connectors will eject the card to a distance of 4.0mm, allowing easy hold and removal of the card.

##### • Exposed termination leads

Easy inspection and rework of the solder termination joints.

#### ◆ DM3CS (Hinge, Push-Pull, manual, without ejection mechanism)

##### • Simple and reliable card insertion

Hinged metal cover provides location and guides the card during the insertion / removal. Closing of the cover confirms the electrical and mechanical connection with a tactile click sensation.

##### • Reliable contact with the card contact pads

Unique contact design and card slide action will clean the contact areas of the card.

##### • Accessible termination areas

Contact solder terminations may be inspected and reworked.

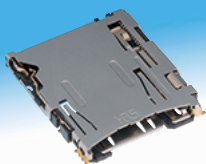
#### ◆ DM3D (Push -Pull, manual, without ejection mechanism)

##### • Partial card insertion hold

Card will not fall-out even when it is not fully inserted. Full insertion and electrical / mechanical connection is confirmed with a distinct tactile feel.

##### • Accessible termination areas

An inner lead system that can be reworked is used in this design. Contact solder terminations may be inspected and reworked.

| Card insertion-ejection             | Series | Image                                                                                 | Page |
|-------------------------------------|--------|---------------------------------------------------------------------------------------|------|
| Push-Push                           | DM3AT  |  | 2~4  |
|                                     | DM3BT  |  | 5~6  |
| Hinge-manual insertion/ejection     | DM3CS  |  | 7~8  |
| Push-Pull manual insertion/ejection | DM3D   |  | 9~10 |

## ■Product Specifications (DM3 Series)

|         |                                                   |                                                                                                          |                                                             |
|---------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|
| Ratings | Current rating : 0.5A<br>Voltage rating : 125V AC | Operating temperature range : -25℃ to +85℃ (Note 1)<br>Storage temperature range : -40℃ to +85℃ (Note 2) | Operating humidity range : RH 95% max.<br>(No condensation) |
|---------|---------------------------------------------------|----------------------------------------------------------------------------------------------------------|-------------------------------------------------------------|

| Item                            | Specification                                                                                                                               | Conditions                                                                                                       |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| 1. Insulation resistance        | 1000MΩ min. (Initial value)                                                                                                                 | Measure at 500V DC                                                                                               |
| 2. Withstanding voltage         | No flashover or insulation breakdown                                                                                                        | 500 V AC / 1 minute                                                                                              |
| 3. Contact resistance           | 100mΩ max. (Initial value)                                                                                                                  | 1mA                                                                                                              |
| 4. Vibration                    | No electrical discontinuity of 100 ns or longer<br>No damage, cracks or parts dislocation.                                                  | Frequency : 10 to 55Hz, single amplitude of 0.75mm,<br>3 directions for 2 hours                                  |
| 5. Humidity                     | Contact resistance : 40mΩ max. (change from initial value)<br>Insulation resistance : 100MΩ min.<br>No damage, cracks or parts dislocation. | 96 hours at of 40 ± 2℃, and humidity of 90 to 95%                                                                |
| 6. Temperature cycle            | Contact resistance : 40mΩ max. (change from initial value)<br>Insulation resistance : 100MΩ min.<br>No damage, cracks or parts dislocation. | -55℃ → 5 to 35℃ → 85℃ → 5 to 35℃<br>Times : 30 min. → 5 min. → 30 min. → 5 min.<br>5 cycles                      |
| 7. Durability                   | Contact resistance : 40mΩ max. (change from initial value)                                                                                  | 10,000 cycles, 400 to 600 cycles per hour (DM3AT, DM3B)<br>5,000 cycles, 400 to 600 cycles per hour (DM3C, DM3D) |
| 8. Resistance to soldering heat | No deformation of components affecting performance.                                                                                         | Reflow : At the recommended temperature profile<br>Manual soldering : 350℃ for 3 seconds                         |

Note 1 : Includes temperature rise caused by current flow.  
Note 2 : The term "storage" refers to products stored for long period prior to mounting and use.

## ■Materials / Finish

### DM3AT, DM3BT

| Part             | Material                                             | Finish                                                | Remarks |
|------------------|------------------------------------------------------|-------------------------------------------------------|---------|
| Insulator        | LCP                                                  | Color : Black                                         | UL94V-0 |
| Contacts         | Copper alloy                                         | Contact area : Gold plated<br>Lead area : Gold plated | _____   |
| Guide cover      | Stainless steel (DM3AT)<br>Copper alloy (DM3BT)      | Lead area : Gold plated                               | _____   |
| Other components | Stainless steel (DM3AT, DM3BT)<br>Piano wire (DM3BT) | _____ Nickel plated                                   | _____   |

### DM3CS, DM3D

| Part        | Material        | Finish                                                | Remarks |
|-------------|-----------------|-------------------------------------------------------|---------|
| Insulator   | LCP             | Color : Black                                         | UL94V-0 |
| Contacts    | Copper alloy    | Contact area : Gold plated<br>Lead area : Gold plated | _____   |
| Guide cover | Stainless steel | _____ (DM3CS)<br>Tin plated (DM3D)                    | _____   |

## ■Product Number Structure

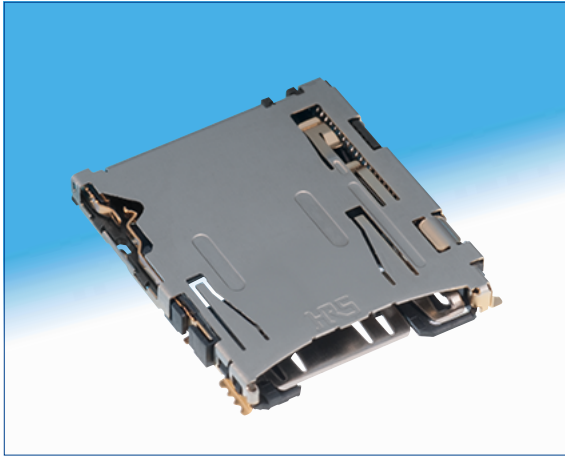
Refer to the chart below when determining the product specifications from the product number.  
Please select from the product numbers listed in this catalog when placing orders.

**DM3 AT – SF – PEJM5**

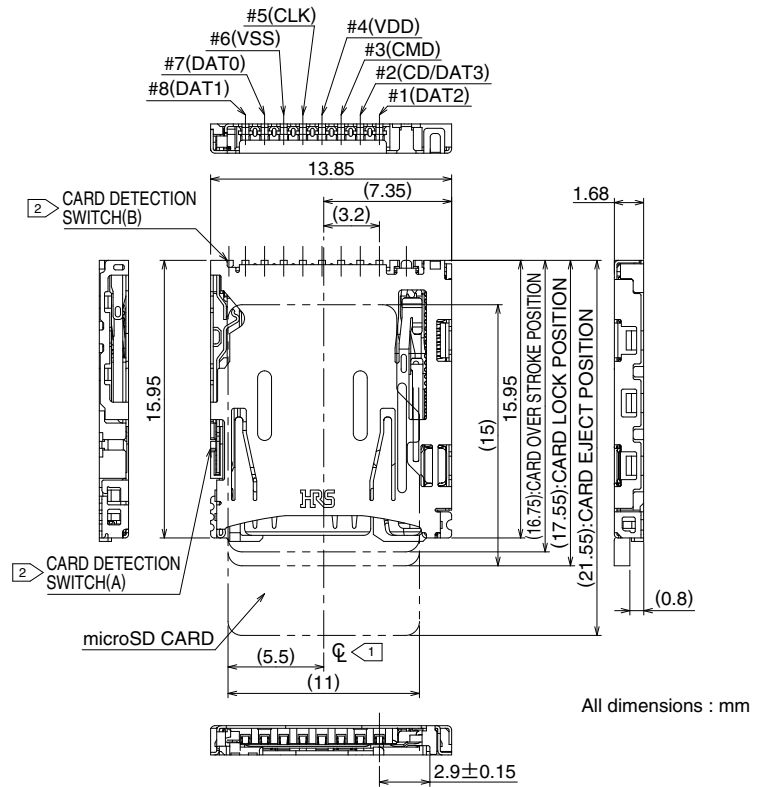
①      ②      ③      ④

|                                                                                                                                                                                                                                                                                                                                         |                                                                                                            |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| ① Series name : DM3                                                                                                                                                                                                                                                                                                                     | ③ Termination type : SF Right-angle SMT(Standard)<br>DSF Right-angle SMT(Reverse)                          |
| ② Connector type : AT Push-Push (ejection mechanism), Top board mounting (Standard)<br>BT Push-Push (ejection mechanism), Bottom board mounting (Reverse)<br>CS Hinge, Push-Pull (no ejection mechanism), Top board mounting (Standard)<br>D Push-Pull (no ejection mechanism), Top board mounting (Standard)<br>Number of contacts : 8 | ④ Card ejection code : PEJM5, PEJS<br>(Push insert/push eject)<br>None : Manual card<br>insertion/ejection |

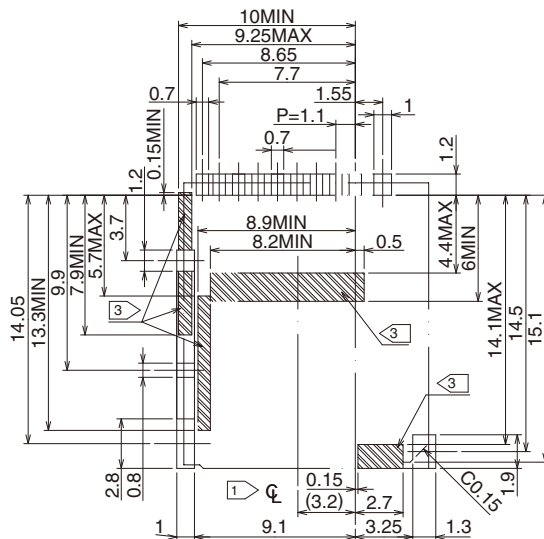
## DM3AT Push-Push (ejection mechanism), Top board mounting (Standard)



| Part No.       | HRS No.    |
|----------------|------------|
| DM3AT-SF-PEJM5 | 609-0031-0 |



## Recommended PCB mounting pattern



Note 1  $\phi$  indicates the center line of the microSD card slot.

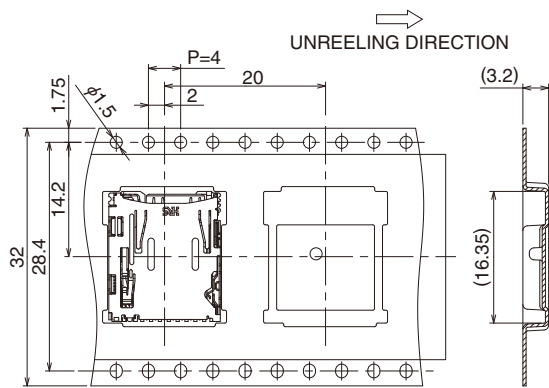
| Card detection switch |     |               |     |
|-----------------------|-----|---------------|-----|
| Without the card      |     | Card inserted |     |
| Open                  |     | Closed        |     |
| (A)                   | (B) | (A)           | (B) |

3 No conductive traces.

All dimensions : mm

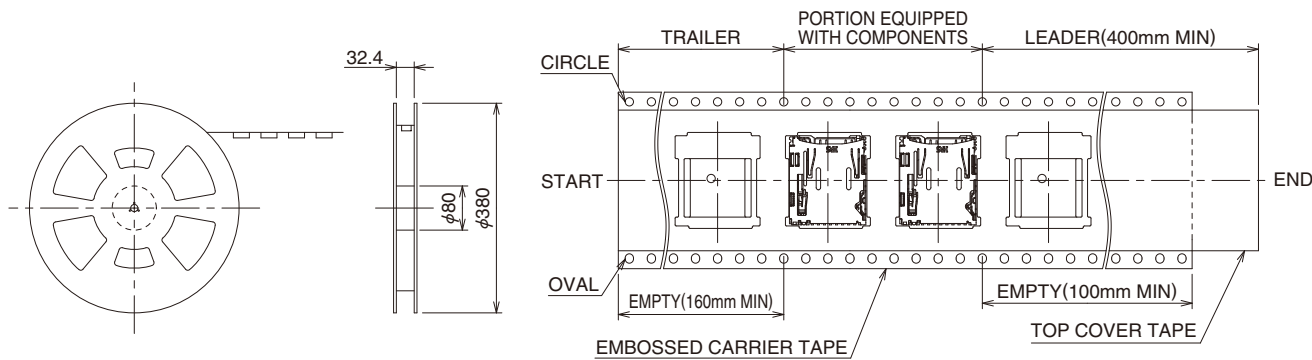
◆Packaging Specifications

- Embossed carrier tape dimensions (1,500 pcs/reel)

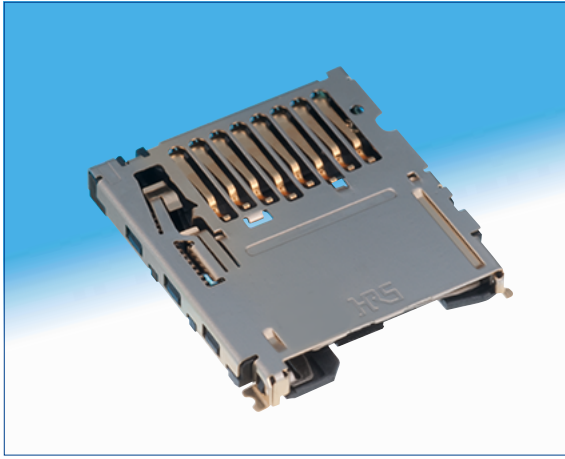


All dimensions : mm

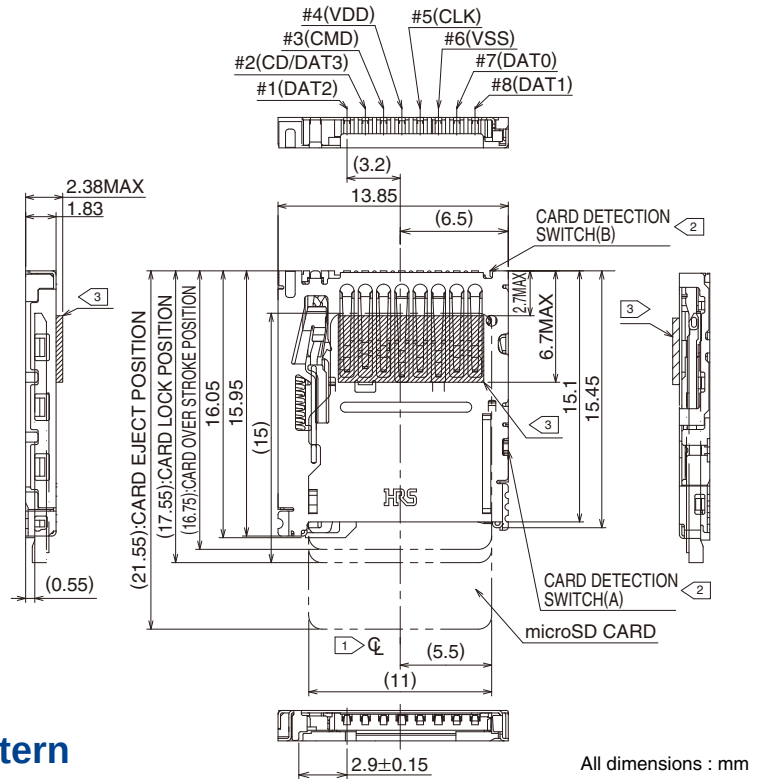
- Reel Dimensions



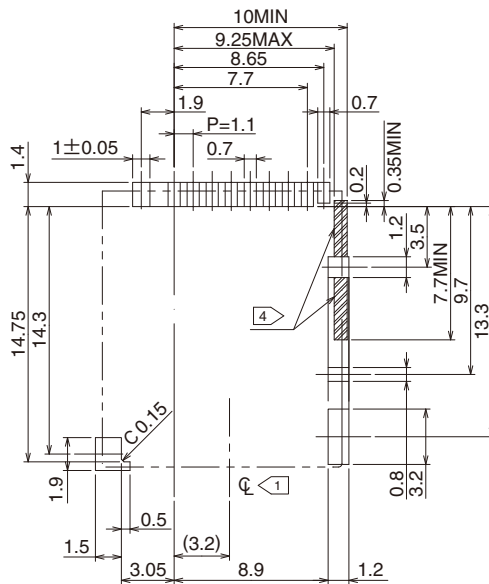
## DM3BT, Push-Push (ejection mechanism), Bottom board mounting (Reverse)



| Part No.       | HRS No.    |
|----------------|------------|
| DM3BT-DSF-PEJS | 609-0029-9 |



## Recommended PCB mounting pattern



Note 1  $\Phi$  indicates the center line of the microSD card slot.

| Card detection switch |     |               |     |
|-----------------------|-----|---------------|-----|
| Without the card      |     | Card inserted |     |
| Open                  |     | Closed        |     |
| (A)                   | (B) | (A)           | (B) |

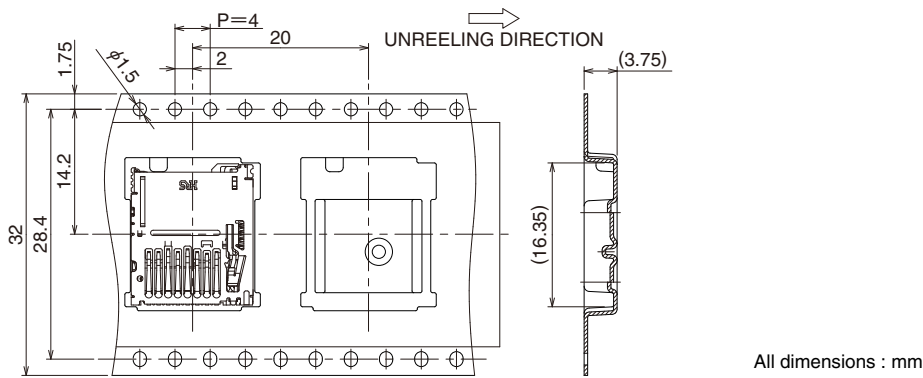
3 Oblique-hatched area is projection of contact.

4  No conductive traces.

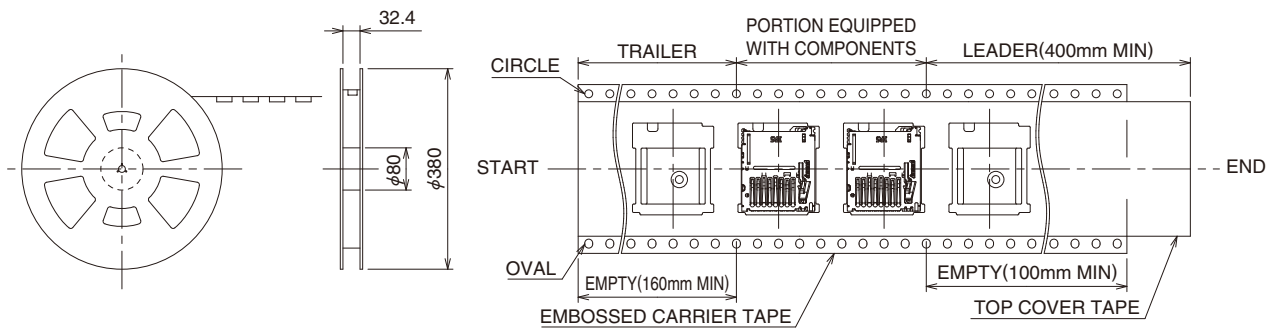
All dimensions : mm

◆Packaging Specifications

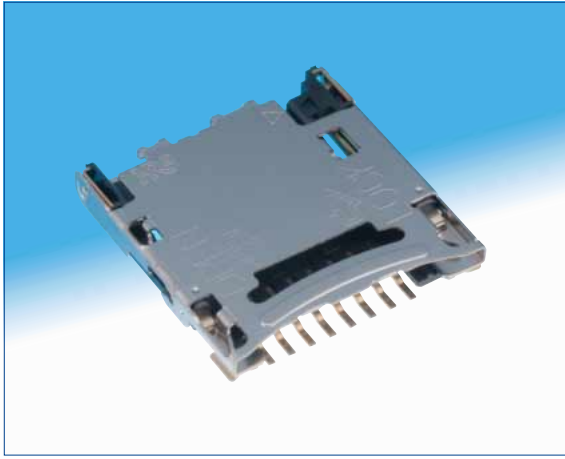
- Embossed carrier tape dimensions (1,200 pcs/reel)



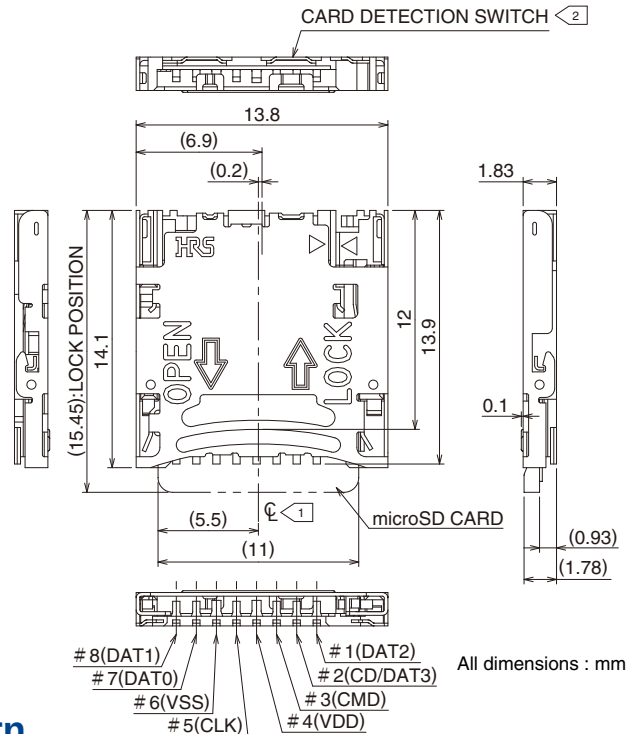
- Reel Dimensions



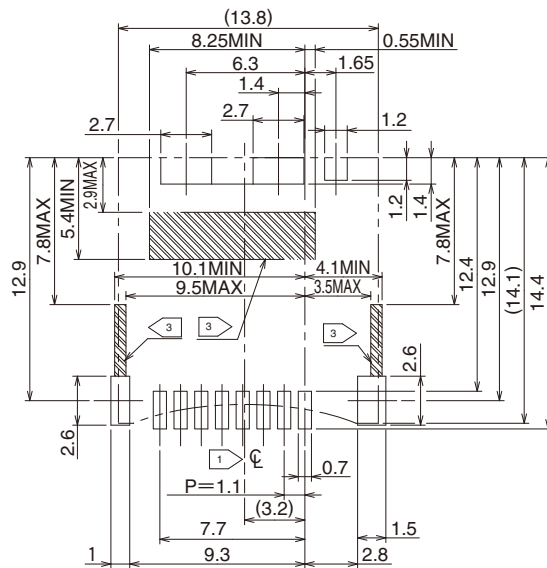
## DM3CS, Hinge, Push -Pull (no ejection mechanism), Top board mounting (Standard)



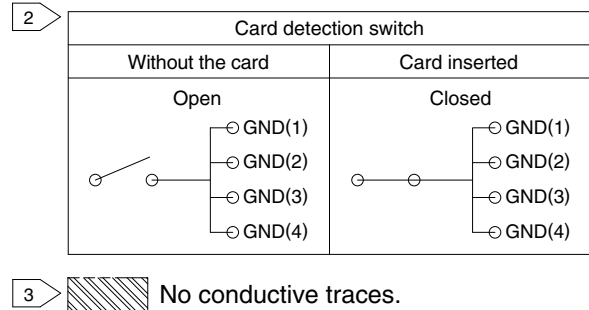
| Part No. | HRS No.    |
|----------|------------|
| DM3CS-SF | 609-0032-3 |



## Recommended PCB mounting pattern



Note 1  $\phi$  indicates the center line of the microSD card slot.



- Embossed carrier tape dimensions (1,300 pcs/reel)



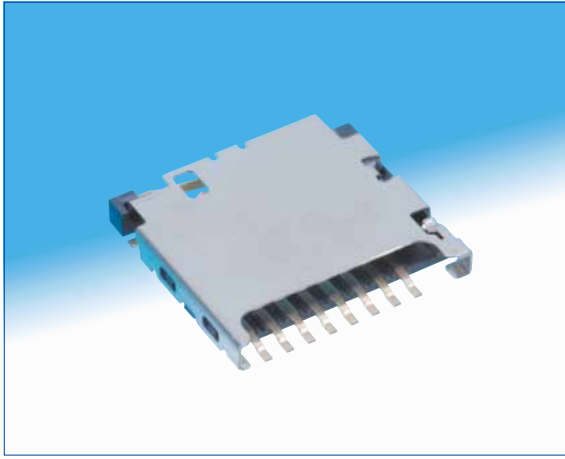
Technical drawing of the 35mm film magazine, showing three views: top, side, and front.

**Top View:** A circular view showing the internal layout of the film magazine. The outer diameter is  $\phi 380$  mm. The inner diameter of the central hub is  $\phi 80$  mm. The distance from the center to the edge of the film gate is 32.4 mm. The film gate is divided into four quadrants by a crosshair.

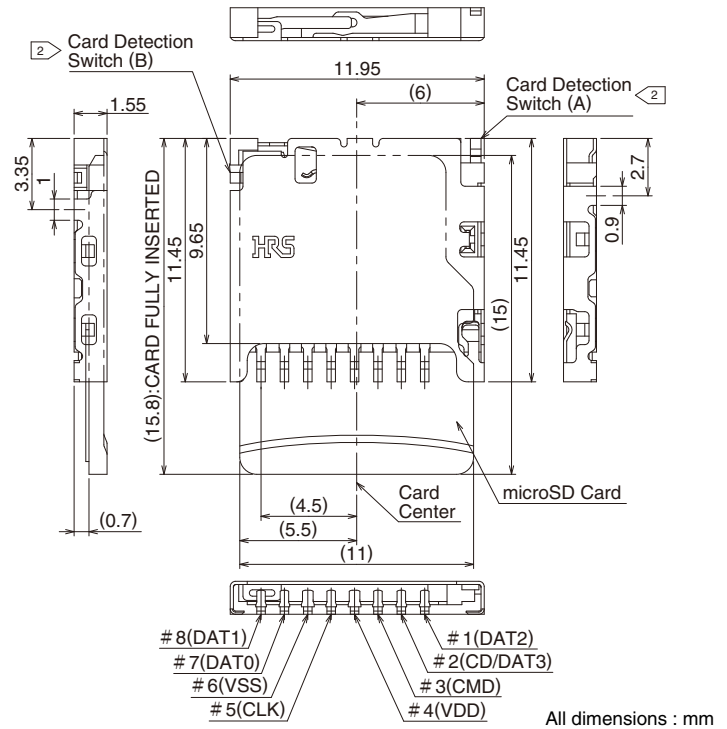
**Side View:** A cross-sectional view showing the internal components. The total height is 38 mm. The height of the central hub is 80 mm. The height of the film gate is 32.4 mm. The distance from the center to the edge of the film gate is 32.4 mm.

**Front View:** A detailed view of the film magazine showing the internal components. The top cover tape is labeled "EMBOSSED CARRIER TAPE". The bottom cover tape is labeled "TOP COVER TAPE". The film magazine is divided into three main sections: "TRAILER", "PORTION EQUIPPED WITH COMPONENTS", and "LEADER (400mm MIN)". The "TRAILER" section is labeled "EMPTY (160mm MIN)". The "PORTION EQUIPPED WITH COMPONENTS" section is labeled "EMPTY (100mm MIN)". The "LEADER" section is labeled "EMPTY (100mm MIN)". The film magazine is labeled "CIRCLE", "START", and "OVAL". The "START" and "OVAL" labels are positioned on the left side of the film magazine. The "END" label is positioned on the right side of the film magazine.

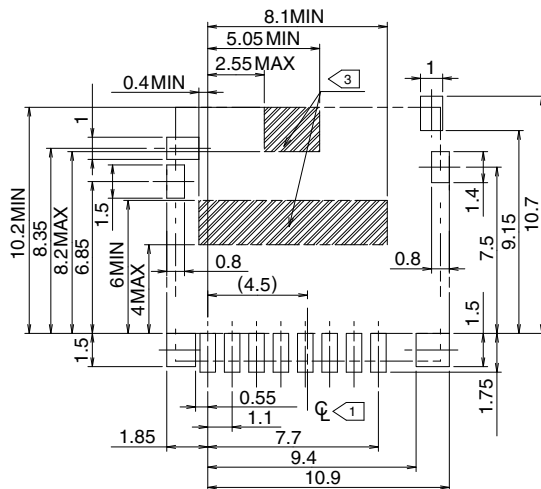
## DM3D, Push-Pull (no ejection mechanism), Top board mounting (Standard)



| Part No. | HRS No.    |
|----------|------------|
| DM3D-SF  | 609-0025-8 |



## Recommended PCB mounting pattern



Note 1  $\Phi$  indicates the center line of the microSD card slot.

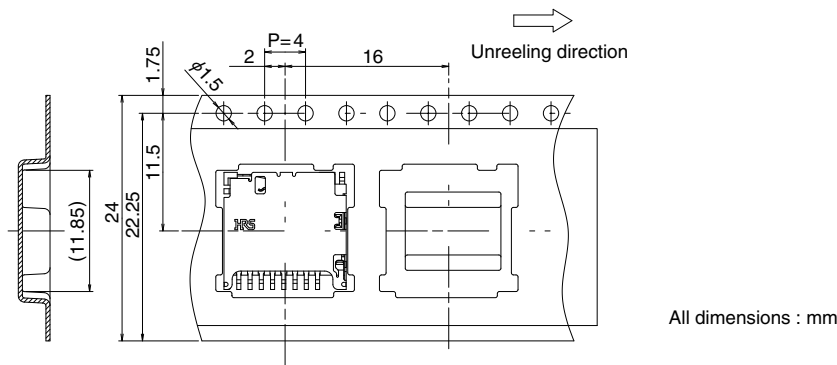
| Card detection switch |     |               |     |
|-----------------------|-----|---------------|-----|
| Without the card      |     | Card inserted |     |
| Open                  |     | Closed        |     |
| (A)                   | (B) | (A)           | (B) |

3 No conductive traces.

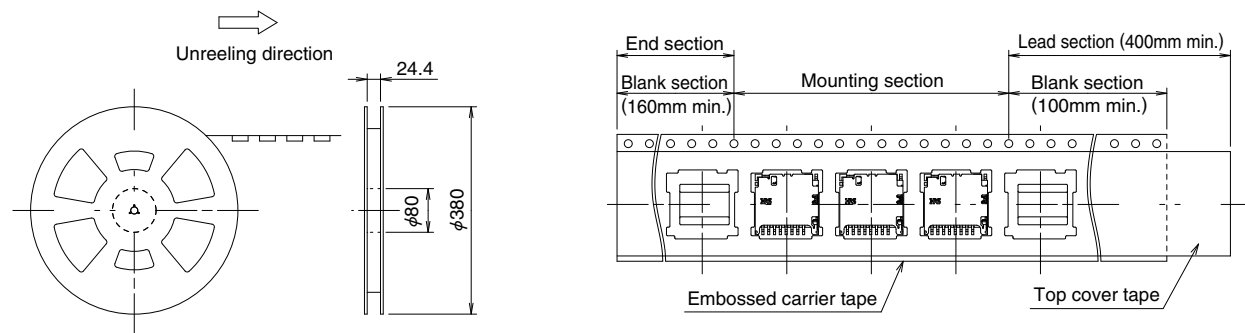
All dimensions : mm

◆Packaging Specifications

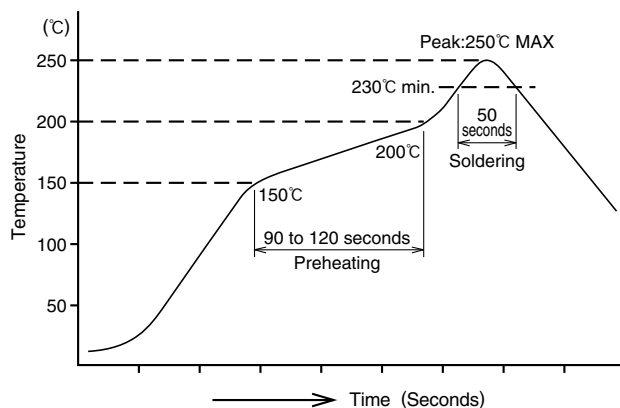
- Embossed carrier tape dimensions (2,000 pcs/reel)



- Reel Dimensions



## ◆ Recommended temperature profile



### HRS test condition

Solder method : Reflow, IR/hot air  
 Environment : Room air  
 Solder composition : Paste, 96.5%Sn/3.0%Ag/0.5%Cu  
 (Senju Metal Industry, Co., Ltd.'s  
 Part Number:M705-GRN360-K2-V)  
 Test board : Glass epoxy 60mm×100mm×1.0mm thick  
 Metal mask : 0.12mm thick  
 Number of reflow cycles : 2cycles max.

The temperature profiles shown are based on the above conditions.

In individual applications the actual temperature may vary, depending on solder paste type, volume / thickness and board size / thickness. Consult your solder paste and equipment manufacturer for specific recommendations.

## ◆ Precautions

1. Do not immerse or clean the entire connector with cleaning solutions as this may affect proper operation of the ejection mechanism and electrical performance of the connector
2. Do not apply excessive force to the connector when handling or after installation on the PC board.
3. The connectors will reliably connect and operate with the correctly inserted microSD™ cards.  
 Follow the correct insertion / ejection procedure for the specific connector in use.  
 Attempts of incorrect insertion of the card may cause damage to the connector or the card.
4. The connector must be correctly mounted on the PC board before the card can be inserted. Do not insert the card in the un-mounted connector.
5. Mounting on the Flexible Printed Circuit (FPC)  
 To assure correct performance it is recommended that a flat reinforcement plate 0.3 mm min. thick be used under the FPC.
6. Small visible residual manufacturing fluids or tooling marks do not affect connector's performance.
7. Repeated insertions and removal of the cards may leave some marks on the card itself. This will have no affect on the connector performance.

### ● Refer to applicable Operation Manual listed below for additional precautions.

| Series       | Operation Manual Number |
|--------------|-------------------------|
| DM3AT Series | ETAD-F0345              |
| DM3BT Series | ETAD-F0324              |
| DM3CS Series | ETAD-F0335              |
| DM3D Series  | ETAD-F0353              |

**USA:**

**HIROSE ELECTRIC (U.S.A.), INC. HEADQUARTERS CHICAGO OFFICE**  
2300 Warrenville Road, Suite 150,  
Downers Grove, IL 60515  
Phone : +1-630-282-6700  
<http://www.hirose.com/us/>

**USA:**

**HIROSE ELECTRIC (U.S.A.), INC. SAN JOSE OFFICE**  
2841 Junction Ave, Suite 200  
San Jose, CA. 95134  
Phone : +1-408-253-9640  
Fax : +1-408-253-9641  
<http://www.hirose.com/us/>

**USA:**

**HIROSE ELECTRIC (U.S.A.), INC. DETROIT OFFICE (AUTOMOTIVE)**  
17197 N. Laurel Park Drive, Suite 253,  
Livonia, MI 48152  
Phone : +1-734-542-9963  
Fax : +1-734-542-9964  
<http://www.hirose.com/us/>

**THE NETHERLANDS:**

**HIROSE ELECTRIC EUROPE B.V.**  
Hogehillweg #8 1101 CC Amsterdam Z-O  
Phone : +31-20-6557460  
Fax : +31-20-6557469  
<http://www.hirose.com/eu/>

**GERMANY:**

**HIROSE ELECTRIC EUROPE B.V. GERMAN BRANCH**  
Schoenbergstr. 20, 73760 ostfildern  
Phone : +49-711-456002-1  
Fax : +49-711-456002-299  
<http://www.hirose.com/eu/>

**GERMANY:**

**HIROSE ELECTRIC EUROPE B.V. NUREMBERG OFFICE**  
Neumeyerstrasse 22-26, 90411 Nurnberg  
Phone : +49-911 32 68 89 63  
Fax : +49-911 32 68 89 69  
<http://www.hirose.com/eu/>

**GERMANY:**

**HIROSE ELECTRIC EUROPE B.V. HANOVER OFFICE**  
Bayernstr. 3, Haus C 30855 Langenhagen, Germany  
Phone : +49-511 97 82 61 30  
Fax : +49-511 97 82 61 35  
<http://www.hirose.com/eu/>

**FRANCE:**

**HIROSE ELECTRIC EUROPE B.V. PARIS OFFICE**  
Regus La Garenne Colombes, Place de La Belgique,  
71 Boulevard National La Garenne Colombes, 92250, France  
Phone : +33 (0) 1 7082 3170  
Fax : +33 (1) 7082 3101  
<http://www.hirose.com/eu/>

**UNITED KINGDOM:**

**HIROSE ELECTRIC EUROPE BV (UK BRANCH)**  
4 Newton Court, Kelvin Drive, Knowlhill,  
Milton Keynes, MK5 8NH  
Phone : +44-1908 202050  
Fax : +44-1908 202058  
<http://www.hirose.com/eu/>

**CHINA:**

**HIROSE ELECTRIC (SHANGHAI) CO., LTD.**  
1601, Henderson Metropolitan, NO.300, East Nanjing  
Road, Huangpu District, Shanghai, China 200001  
Phone : +86-21-6391-3355  
Fax : +86-21-6391-3335  
<http://www.hirose.com/cn/>

**CHINA:**

**HIROSE ELECTRIC (SHANGHAI) CO., LTD. BEIJING BRANCH**  
A1001, Ocean International Center, Building 56# East 4th  
Ring Middle Road, ChaoYang District, Beijing, 100025  
Phone : +86-10-5165-9332  
Fax : +86-10-5908-1381  
<http://www.hirose.com/cn/>

**CHINA:**

**HIROSE ELECTRIC TECHNOLOGIES (SHENZHEN) CO., LTD.**  
Room 09-13, 19/F, Office Tower Shun Hing Square, Di Wang Commercial Centre,  
5002 Shen Nan Dong Road, Shenzhen City, Guangdong Province, 518008  
Phone : +86-755-8207-0851  
Fax : +86-755-8207-0873  
<http://www.hirose.com/cn/>

**HONG KONG:**

**HIROSE ELECTRIC HONGKONG TRADING CO., LTD.**  
Room 1001, West Wing, Tsim Sha Tsui Centre, 66  
Mody Road, Tsim Sha Tsui East, Kowloon, Hong Kong  
Phone : +852-2803-5338  
Fax : +852-2591-6560  
<http://www.hirose.com/hk/>

**TAIWAN:**

**HIROSE ELECTRIC TAIWAN CO., LTD.**  
103 8F, No.87, Zhengzhou Rd., Taipei  
Phone : +886-2-2555-7377  
Fax : +886-2-2555-7350  
<http://www.hirose.com/tw/>

**KOREA:**

**HIROSE KOREA CO., LTD.**  
250, Huimanggongwon-ro, Siheung-si,  
Gyeonggi-do, Korea, 15083  
Phone : +82-31-496-7000 or 7124  
Fax : +82-31-496-7100  
<http://www.hirose.co.kr/>

**SINGAPORE:**

**HIROSE ELECTRIC SINGAPORE PTE. LTD.**  
10 Anson Road #26-16, International Plaza  
079903, Singapore  
Phone : +65-6324-6113  
Fax : +65-6324-6123  
<http://www.hirose.com/sg/>

**INDIA:**

**HIROSE ELECTRIC SINGAPORE PTE. LTD. DELHI LIAISON OFFICE**  
Office NO.552, Regus-Green Boulevard, Level5, Tower C,  
Sec62, Plot B-9A, Block B, Noida, 201301, Uttar Pradesh, India  
Phone : +91-12-660-8018  
Fax : +91-120-4804949  
<http://www.hirose.com/sg/>

**INDIA:**

**HIROSE ELECTRIC SINGAPORE PTE. LTD. BANGALORE LIAISON OFFICE**  
Unit No-403, 4th Floor, No-84, Barton Centre, Mahatma  
Gandhi (MG) Road, Bangalore 560 001, Karnataka, India  
Phone : +91-80-4120 1907  
Fax : +91-80-4120 9908  
<http://www.hirose.com/sg/>

**MALAYSIA:**

**PENANG REPRESENTATIVE OFFICE**  
1-21-01, Suntech @ Penang Cybercity (1164), Lintang  
Mayang Pasir 3, 11950, Bayan Baru, Penang, Malaysia.  
Phone : +604-619-2564  
Fax : +604-619-2574  
<http://www.hirose.com/sg/>

**THAILAND:**

**BANGKOK OFFICE (REPRESENTATIVE OFFICE)**  
Unit 4703, 47th FL., 1 Empire Tower, South Sathorn  
Road, Yannawa, Sathorn, Bangkok 10120 Thailand  
Phone : +66-2-686-1255  
Fax : +66-2-686-3433  
<http://www.hirose.com/sg/>



## HIROSE ELECTRIC CO.,LTD.

2-6-3, Nakagawa Chuoh, Tsuzuki-Ku, Yokohama-Shi 224-8540, JAPAN

TEL: +81-45-620-3526 Fax: +81-45-591-3726

<http://www.hirose.com>

<http://www.hirose-connectors.com>